```python
from sklearn.metrics import mean_absolute_error, mean_squared_error
import numpy as np

mae = mean_absolute_error(test, forecast)
rmse = np.sqrt(mean_squared_error(test, forecast))
mape = np.mean(np.abs((test - forecast) / test)) * 100

accuracy = 100 - mape

print("MAE :", round(mae,2))
print("RMSE:", round(rmse,2))
print("MAPE:", round(mape,2), "%")
print("Accuracy:", round(accuracy,2), "%")
```

```
... MAE : 4.7
RMSE: 5.57
MAPE: 16.84 %
Accuracy: 83.16 %
```

Your ARIMA model evaluation results are:

Metric	Value
MAE	4.70
RMSE	5.57
MAPE	16.84%
Accuracy	83.16%

—--------

1️⃣ MAE (Mean Absolute Error)

Formula:

$$MAE = \frac{1}{n}\sum|Actual - Forecast|$$

Meaning:

MAE measures the average difference between actual and predicted temperatures.

Your Result:

MAE = 4.70

👉 On average, your ARIMA model's prediction differs from the actual temperature by **4.7°C**.

2️⃣ RMSE (Root Mean Squared Error)

Formula:

$$RMSE = \sqrt{\frac{1}{n}\sum(Actual - Forecast)^2}$$

Meaning:
RMSE gives higher penalty to larger prediction errors.

Your Result:
RMSE = 5.57

👉 Large forecasting errors have increased the RMSE to **5.57°C**.

👉 Lower RMSE indicates a better model.

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3️⃣ MAPE (Mean Absolute Percentage Error)

Formula:

$$MAPE = \frac{1}{n}\sum\left|\frac{Actual - Forecast}{Actual}\right| \times 100$$

Meaning:
Shows the prediction error as a percentage.

Your Result:
MAPE = 16.84%

👉 On average, the forecast differs from the actual temperature by **16.84%**.

MAPE Interpretation

MAPE	Performance
<10%	Excellent
10–20%	Good
20–50%	Acceptable
>50%	Poor

✅ Your model falls in the **Good Forecasting Performance** category.

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4️⃣ Accuracy

Formula Used:

$$Accuracy = 100 - MAPE$$

Your Result:

$$100 - 16.84 = 83.16\%$$

👉 The ARIMA model achieved an approximate forecasting accuracy of **83.16%**.

Project Answer:

> "The ARIMA model was evaluated using MAE, RMSE, and MAPE metrics. The model achieved an MAE of 4.70, RMSE of 5.57, and MAPE of 16.84%. Based on MAPE, the forecasting accuracy is approximately 83.16%, indicating good predictive performance for Chennai temperature forecasting."

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📊 **Model Performance**

✅ MAE : **4.70**

✅ RMSE : **5.57**

✅ MAPE : **16.84%**

✅ Forecast Accuracy : **83.16%**

⭐ **Performance Rating: Good Forecasting Model**